MILITARY SPECIFICATION

PROPELLANTS, NITROGEN TETROXIDE

This amendment forms a part of Military Specification MIL-P-26539D, dated 28 Aug 1989, and is approved for use by all Departments and Agencies of the Department of Defense

PAGE 1

1 1· Delete and substitute

"1 1 Purpose This specification covers the requirements for five types and two grades of nitrogen tetroxide ($N_2O_4$) propellants "

1 2: Delete and substitute

"1.2 Classification Nitrogen tetroxide propellants shall be of the following designated types and grades as specified (6 2).

Types

| | | |
|---|---|---|
| NTO | - | Nominal 99 5% $N_2O_4$ purity with red-brown color |
| MON-1 | - | Nominal 99% $N_2O_4$ and 1% NO with green color |
| MON-3 | - | Nominal 97% $N_2O_4$ and 3% NO with green color |
| MON-10 | - | Nominal 90% $N_2O_4$ and 10% NO with green color |
| MON-25 | - | Nominal 75% $N_2O_4$ and 25% NO with green color |

Grades

| | | |
|---|---|---|
| Standard | - | no iron requirement. |
| Low-Iron | - | maximum 0 5 parts per million iron " |

PAGE 2

2.2· Add the following·

"International Civil Air Organization (ICAO)

Technical Instructions for the Safe Transport of Dangerous Goods by Air

International Air Transport Association (IATA)

Dangerous Goods Regulations

International Maritime Organization (IMO)

International Maritime Dangerous Goods Code (IMDG Code)"


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PAGE 4

TABLE I Delete and substitute·

"TABLE I. Chemical Composition and Physical Properties

| Composition | NTO (Red-Brown) | MON-1 (Green) | MON-3 (Green) | MON-10 (Green) | MON-25 (Green) | Test Para |
|---|---|---|---|---|---|---|
| $N_2O_4$ assay % wt, MIN | 99.5 | --- | --- | --- | --- | 4 5 3 |
| NO content % wt, MAX | | 1 0 | 3 0 | 11.0 | 26.0 | 4 5.2 |
| % wt, MIN | 1/ | 0 6 | 2.5 | 10 0 | 25.0 | |
| $N_2O_4$ + NO % wt, MIN | --- | 99 5 | 99.5 | 99.5 | 99 5 | 4 5.3 |
| Water equivalent % wt, MAX | 0 17 | 0.17 | 0.17 | 0 17 | 0.17 | 4 5 4 |
| Chloride Content % wt, MAX | 0.040 2/ | 0.040 | 0 040 | 0 040 | 0 040 | 4 5 5 |
| Non-Volatile Residue 3/ mg/l, MAX | --- | 10 0 | 10 0 | 10.0 | 10 0 | 4 5 6 |
| Iron Content 3/ ppm wt, MAX | --- | 0 5 | 0.5 | 0 5 | 0 5 | 4 5.7 |
| Particulate mg/l, MAX | 10.0 | 10.0 | 10 0 | 10 0 | 10.0 | 4.5.8 |

1/ The NO content shall be limited to that which does not change the specified red-brown color of the propellant (3 4)

2/ This test need not be performed on NTO propellant if the material was manufactured by the ammonia-oxidation process

3/ This requirement applies to the low-iron grade of the propellant only."



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PAGE 6

4.4.2.1.3: Add the following sentence:

"Each cylinder of low-iron grade material shall constitute a separate lot and shall be sampled."

PAGE 8

4.5.2.1.3, delete the calculation for NO and substitute:

$$\text{"NO, percent by weight} = \frac{[y\text{-}0\ 003857\ (z - \frac{x+y}{1.49})]\ 187\ 5}{x} - 0.15\text{"}$$

PAGE 15

4 5.3.2, line 15: Delete "methyl red or methyl orange" and subsitute "bromothymol blue".

PAGE 16

4.5.3.4 a(1), line 3: Delete "methyl red or methyl orange" and subsitute "bromothymol blue".

4.5.3.4 a(3) Delete and subsitute:

"Bromothymol blue indicator solution: 0.1 grams in 16 ml of 0.01 N NaOH and 234 ml of distilled water."

4.5.3.4 a(5), line 2: Delete "methyl red" and substitute "bromothymol blue".

PAGE 21

4.5.4.2.1: Add the following sentence:

"Throughout the procedure, minimize the exposure of the oxidized $N_2O_4$ to the atmosphere."

PAGE 22

4.5.4 2.4 f, line 1: Add ", or equivalent." to the end of the line.



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PAGE 23

4 5 5 1 1, line 13 Delete "a pH of 2 5 to 3 0 is reached" and substitute "the pH stabilizes between 2 5 to 3 0"

PAGE 26

4 5 6 1 Delete and substitute

"4 5 6 1 Procedure. Thoroughly clean two platinum evaporating dishes per sample and one for the blank Dry the dishes in an oven at 220 ± 10°F (105 ± 5°C) After drying cover the dishes with a watch glass and allow them to cool next to the balance for 30 minutes Weigh each dish to the nearest 0 01 milligram Add 1 ml concentrated $H_2SO_4$ to one of the dishes Evaporate the $H_2SO_4$ by drying in an oven for one hour beyond the point when no liquid remains in the dish This dish will be referred to as the blank. Place a hot plate in an exhaust hood and place the remaining weighed dishes on it Add 1 ml concentrated $H_2SO_4$ to each dish. Transfer 50 ml of the $N_2O_4$ to each dish Using the hot plate, evaporate the sample until only 1 to 2 ml remain in each dish Oven dry the dishes for one hour beyond the point when no liquid remains in each dish. These dishes will be referred to as the sample dish After drying cover the dishes with a watch glass and allow to cool next to the balance for 30 minutes Weigh each dish to the nearest 01 mg "

4 5 6 2 Delete and substitute.

"4 5 6 2 Calculation The mg/l non-volatile residue shall be calculated by the following formula

$$NVR\ (mg/l) = \frac{(W_S - W_E)}{V_S} - \frac{W_B - W_E}{V_B}$$

where

$W_S$ = Weight of sample dish after drying (mg)

$W_E$ = Weight of empty dish (mg)

$W_B$ = Weight of blank dish after drying (mg)

$V_S$ = Volume of sample used (liters)

$V_B$ = Volume of blank used (liters)"



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PAGE 27

4 5 7, line 1: Delete "MON-1 and the MON-3" and substitute "propellant"

4 5.7 1, line 1 Delete "Using a 100 ml burette"

4.5.7 1, line 2: Delete "MON-1 or MON-3" and substitute "propellant"

4 5 7.2, line 1. Delete "MON-1 or MON-3" and substitute "propellant".

PAGE 32

5.1: Delete and substitute:

"The product furnished under this specification is a hazardous material Packaging shall comply with Code of Federal Regulation 49 (49 CFR), DOT Special Permit/Exemption, and United Nations (UN)/International requirements as stipulated in ICAO, IATA, and IMO, as applicable for mode of transportation and destination "

PAGE 33

5 4. Delete and substitute.

"Each container shall be labeled, marked, and placarded in accordance with 49 CFR, DOT Special Permit/Exemption, MIL-STD-129, and UN/International requirements as stipulated in ICAO, IATA, and IMO, as applicable for mode of transportation and destination. In addition, each container shall be marked "INHALATION HAZARD" An identification tag, precautionary label, and container color code with name identification shall be used on each cylinder."

6.2.1, line 1 Delete "composition" and substitute "composition, grade".

PAGE 34

6 3e, line 1. Delete "MON-1 and MON-3 product" and substitute "propellant"

Custodians
Army - MI
Navy - AS
Air Force - 68

Review Activities
Air Force - 12, 14, 19

Preparing Activity
Air Force - 68

Civil Agency Interest
NASA

(Project 9135-0124)



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